

UNDERSPECIFIED PLAN

```
sequential([
  a(PickUpAction)(
    arm=LEFT,
    object_designator=
```

```
    a(DetectedMontessoriShape)(
      category=CUBE)
    .where(
      Colored(shape, CYAN),
      SupportedBy(shape, lid)),
```

```
    grasp_description=
```

```
    a(GraspDescription)(
      approach_direction=...,
      vertical_alignment=TOP,
      end_effector=LEFT_HAND)
```

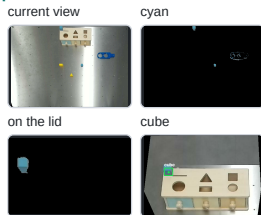
```
  ),
  an(InsertionAction)(
    arm=LEFT,
    object_designator=shape,
    grasp_description=grasp,
    target=
```

```
    a(ShapeSortingHole)(
      shape_category=...)
    .from_(board.apertures)
```

```
  ])
]
```

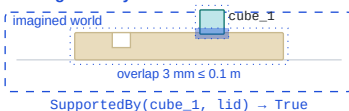
RESOLVED BY

PerceptionBackend

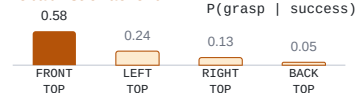


↓ spawn

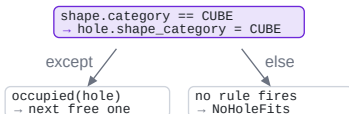
Working memory backend



ProbabilisticBackend



Ripple-down rules



RESOLVED PLAN

```
sequential([
  PickUpAction(
    arm=LEFT,
    object_designator=
```

```
    cube_1 # CYAN, CUBE
    at (0.72, 0.13, 0.10) m,
```

```
    SupportedBy(cube_1, lid) ✓
```

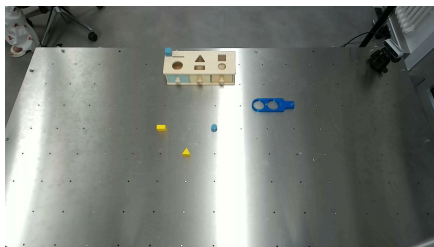
```
    grasp_description=
```

```
    grasp_1 = GraspDescription(
      FRONT, TOP,
      LEFT_HAND)
```

```
  ),
  InsertionAction(
    arm=LEFT,
    object_designator=cube_1,
    grasp_description=grasp_1,
    target=
```

```
    ShapeSortingHole(
      shape_category=CUBE)
```

```
  ])
]
```



Robot

executes the resolved plan

